

Jaypee University of Engineering & Technology, Guna**T-1 (Even Semester 2022)****18B11CI412 –Algorithms and Problem Solving**

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 5 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

Q1. Suppose there are N Jobs in a queue to be done, and each job has its own priority. The job with maximum priority will get completed first than others. At each instance we are completing a job with maximum priority and at the same time we are also interested in inserting a new job in the queue with its own priority. So at each instance we have to check for the job with given priority to complete it and also insert if there is a new job. This task can be very easily executed using a heap by considering N jobs as N nodes of the tree. Write an efficient algorithm to delete a particular job from the data set after completion. (Job index given by user). [03]

Q2. The recurrence relation $T(n) = 7T(n/2) + n^2$ describe the running time of an algorithm A . The algorithm A competes with algorithm X which has a running time of $T(n) = aT(n/4) + n^2$. What is the largest integer value for a such that algorithm X is asymptotically faster than algorithm A ? [03]

Q3. Consider two algorithms A and B that takes time in $O(n^2)$ and $O(n^3)$, respectively. Could there exist an implementation of algorithm B that would be more efficient (in terms of computing time) than an implementation of algorithm A on all instances? Justify your answer. [02]

(b) Explain why the statement, "The running time of algorithm A is at least $O(n^2)$," is meaningless. [01]

Q4. Prove that if $f(n) \in O(n)$ then $[f(n)]^2 \in O(n^2)$. [02]

- (b) Consider an algorithm that takes time in $\Theta(n^{\log 3})$ to solve instances of size n . Is it correct [01]
to say that it takes $O(n^{1.59})$ and $\Omega(n^{1.59})$ in $\Theta(n^{1.59})$? Justify your answers.

Q5. (a) Solve given recurrence relation using recursion tree method. [02]

$T(n) = T(n/4) + T(n/2) + c n^2$

- (b) How much time does insertion sorting take to sort n distinct items in the best case? State [01]
your answer in asymptotic notation.

Semester IV (B.Tech)

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Er. No. 2018041
Academic Year: 2021-22

Jaypee University of Engineering & Technology, Guna

T-1 (Even Semester 2022)
18B11HS411- LIFE SKILLS

Maximum Marks: 15

Maximum duration: 1 Hour

Notes:

1. This question paper has 03 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (except Er. No.).

- Q1. State the need and importance of life skills. Enlist the ten core life skills recommended by WHO. Differentiate between skills and abilities with the help of an example. [2+2+1]
- Q2. Explain Maslow's hierarchy of needs theory in brief with the help of a neat and labeled diagram. Analyze the application of Maslow's theory to an African or Caribbean nation where more than a quarter of the population is unemployed. Mention the grounds on which the ERG theory modifies the Maslow's theory. [2+1.5+1.5]
- Q3. Which kinds of perception are formed based on an individual's comparison of ratio of their job inputs and outcomes with those of others? Briefly give an example of an equity/inequity that a manager of a small business might experience. How would the manager strive to attain equity in the situation that you describe? [2+1.5+1.5]

Jaypee University of Engineering & Technology, Guna
T-1 (Even Semester 2022)
18B11CI411 – COMPUTER NETWORKS

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has **five** questions.
2. Write relevant answers only.
3. Do not write anything on question paper (Except your Er. No.).

- | | Marks |
|---|--------------|
| Q1. (a) An image is 1024×768 pixels with 3 bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it?
(i) Over a 10 Mbps,
(ii) Over a 100 Mbps. | [02] |
| (b) What are the three criteria necessary for an effective and efficient network? | [01] |
| Q2. (a) Explain the three transmission modes in detail. | [01] |
| (b) For n devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology? | [02] |
| Q3. (a) A signal has passed through three cascaded amplifiers, each with a 5dB gain. What is the total gain? How much is the signal amplified? | [02] |
| (b) Name the three types of transmission impairments? | [01] |
| Q4. Summarize the functions of each layer in OSI reference model. | [03] |
| Q5. (a) How many FM radio stations can be allocated in India, if each FM radio station is assigned 200 kHz bandwidth?
(i) With guard band of 100 kHz.
(ii) Without any guard band. | [02] |
| (b) Differentiate between physical and logical addressing utilize in TCP/IP model. | [01] |

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Semester IV (B.Tech)

Er. No. 201804
Academic Year: 2021-22

Jaypee University of Engineering & Technology, Guna

T-1 (Even Semester 2022)

18B11CI413 -OPERATING SYSTEMS

Maximum Duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has three questions.
2. Write relevant answers only.
3. Do not write anything on a question paper (Except your Er. No.).

Marks
[03]

Q1.

Explain the output of a given program and justify the sequence of output lines.

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
int main (int argc, char *argv[])
{
    printf ("Hello world (pid: %d)\n", (int) getpid());
    int rc=fork ();
    if(rc<0) {
        fprintf (stderr, "fork Failed\n");
        exit (1);
    } else if (rc==0) {
        printf ("hello, I am child (pid: %d) \n", (int) getpid());
    } else {
        printf ("hello, I am parent of %d (pid: %d) \n", rc, (int) getpid ());
    }
    return 0;
}
```

Q2.

Suppose that a process is in "RUNNING" state and suddenly a request for some input from the keyboard arrives before the service is completed. Where will the process go? Explain the scenarios by process state diagram.

[04]

Q3.

Consider a set of 5 processes whose arrival time, CPU time needed, and the priorities are given below. Answer the following questions with appropriate calculations and Gantt chart.

Process	Arrival Time (in ms)	Burst Time (in ms)	Priority
P1	0	10	5
P2	2	5	4
P3	4	3	2
P4	7	18	3
P5	8	2	1

(a) If the CPU scheduling policy is SJF (Preemptive) then calculate the average waiting time and response time for process P5. [04]

(b) If the CPU scheduling policy is priority scheduling with preemption, then calculate the average turnaround time and throughput. [04]

Jaypee University of Engineering & Technology, Guna

Er. No. 201804)
Academic Year: 2021-22

T-1 (Even Semester 2022)

18B14HS441-Concept of Digital Marketing

Maximum duration: 1 Hour

Maximum Marks: 15

Notes:

1. This question paper has 3 questions.
2. Write relevant answers only.
3. Do not write anything on question paper (except Enroll No.)

- Q1. (a) 'Digital marketing is extremely important in present scenario. It helps focus on targeted audiences with a global reach, unlike traditional marketing, bringing in high revenue with little investment'. Would you argue that this statement is largely true or false? Why or why not? Use fundamental objectives of digital marketing to justify your argument. [02]
- (b) 'As a part of a positioning strategy, niche marketing may be used as a deliberate marketing strategy to create "safe havens" and business opportunities for many companies (Dalgic & Leeuw, 1994)'. Are you agree with the statement? Justify your case with the help of suitable example. [02]
- Q2. (a) 'Lead generation is a process that builds visibility, credibility, trust, and interest from a specific group of people (potential leads)'. In view of the statement, describe the various strategies to generate potential leads from digital marketing. [03]
- (b) What do you understand by interactivity in digital marketing? Explain different modes used by organizations to interact with stakeholders. [02]
- Q3. (a) 'A competitive environment is a system where different businesses compete with each other by using various marketing channels, promotional strategies, pricing methods, etc'. In view of the statement, examine and discuss the competitive environment framework. [03]
- (b) 'A search engine algorithm is a complex algorithm used by search engines such as Google, Yahoo, and Bing to determine a web page's significance'. Explain ranking algorithm, demonstrate ranking algorithm with the help of an example. [03]

Semester IV (B.Tech)

Er. No. 2612041

Academic Year: 2021-22

Jaypee University of Engineering & Technology, Guna**T-1(Even Semester 2022)****14B11CI711–Artificial Intelligence & Application**

Maximum Duration: 1Hour

Maximum Marks: 15

Notes:

1. This question paper has 4 questions.
2. Write relevant answers only.
3. Do not write anything on question paper.
4. Kindly answer the questions in serial order.

Q1. Compare and contrast the four goals of Artificial Intelligence. **Marks [03]**

Q2. The initial and goal configuration of the 8 puzzle problem is as shown below. Compute and show all the possible heuristics for the following. **[04]**

1	2	3
7	8	4
6		5

Initial State

1	2	3
8		4
7	6	5

Goal State

Q3. Draw all possible state transitions in the Towers of Hanoi puzzle with 3 disks and also mention the shortest solution. **[04]**

Q4. Describe backward chaining. A Rule base contains the following Rule set: **[04]**

Rule 1: If A and C Then F

Rule 2: If A and E Then G

Rule 3: If B Then E

Rule 4: If G Then D

Prove using backward chaining the following condition

If A and B true Then D is true